

Trends in Computer Science

Short Study Notes

CS SE 1109 | Saegis Campus | Sessions 01 · 02 · 03

Session	Topic
01	Introduction to Emerging Technologies
02	Introduction to Artificial Intelligence
03	Internet of Things & Edge Computing

SESSION 01 — Introduction to Emerging Technologies

1.1 What Are Emerging Technologies?

Emerging technologies are typically new innovations — or older technologies finding entirely new applications. They are distinguished by four core characteristics:

Radical Novelty	Fulfil functions using different principles from existing technologies.
Relatively Fast Growth	Rapid adoption, development, and commercialisation.
Coherence	Consistent and structured progress over time — not random bursts.
Prominent Impact	Significant and visible social or economic effects.

1.2 Why Study Emerging Trends?

- **Career Relevance:** The IT job market evolves rapidly. Staying updated keeps you competitive and employable.
- **Innovation:** Understanding trends lets you identify gaps, propose novel solutions, and drive advancement.
- **Decision Making:** Future leaders must evaluate which technologies to adopt — weighing costs, benefits, and ethics.

1.3 The Gartner Hype Cycle

A graphical model representing the **maturity, adoption, and social application** of specific technologies. It helps stakeholders understand where a technology sits in its lifecycle.

1. Innovation Trigger	Early proof-of-concept. No usable product yet. Significant press interest.
2. Peak of Inflated Expectations	Early success stories generate excessive enthusiasm; failures also occur.
3. Trough of Disillusionment	Interest wanes. Experiments fail. Producers shake out.
4. Slope of Enlightenment	2nd/3rd gen products appear. More enterprises adopt.
5. Plateau of Productivity	Mainstream adoption. Market applicability is clear.

Key insight: Not all technologies survive the Trough of Disillusionment — only those that deliver real value reach the Plateau of Productivity.

1.4 Key Technology Themes for 2024/2025

Theme 1 — Artificial Intelligence

Moving beyond Generative AI (ChatGPT) to **Autonomous AI** — agents that plan and execute goals without human intervention. **Machine Customers** are AI algorithms that negotiate and purchase goods.

Theme 2 — IoT & Connectivity

The IoT is evolving into the **Internet of Everything**. **Edge Computing** processes data near sensors to reduce latency. **5G/6G** enables massive machine-type communications and real-time control.

Theme 3 — Immersive Experiences (Spatial Computing)

Merging physical and digital worlds. Includes **AR/VR** for training and collaboration, **Digital Twins** (virtual replicas of physical systems), and **Total Experience (TX)** — unifying customer, employee, and user experience.

Theme 4 — Trust & Security

AI TRiSM (AI Trust, Risk & Security Management) ensures AI is fair, reliable, and secure. **Cybersecurity Mesh** is a flexible composable architecture integrating distributed security services. **Privacy-Enhancing Computation** analyses data while keeping it encrypted.

1.5 Socioeconomic Impact

✓ Positive Impacts	✗ Negative Impacts
Efficiency — automation reduces costs and errors	Job Displacement — automation of routine tasks
Healthcare — AI diagnostics and personalised medicine	Digital Divide — gap between tech haves and have-nots
Connectivity — global access to info and education	Privacy Loss — surveillance capitalism, data breaches
New Jobs — drone pilots, AI prompt engineers, etc.	Dependency — over-reliance on complex systems

Ethical Question to Always Ask: *"Just because we CAN build it, does that mean we SHOULD?"*

SESSION 02 — Introduction to Artificial Intelligence

2.1 What Is Artificial Intelligence?

Artificial Intelligence (AI) is the **simulation of human intelligence processes by machines** — especially computers. These processes include learning, reasoning, problem-solving, perception, and language understanding.

Types of AI by capability:

Narrow AI (Weak AI)	Designed for ONE specific task. Most AI today falls here. Examples: Siri, chess engines, image classifiers.
General AI (AGI)	Hypothetical AI with human-level understanding across all domains. Does not yet exist.
Super AI	Hypothetical AI surpassing human intelligence in all areas. Purely theoretical.

2.2 Machine Learning (ML)

ML is a **subset of AI** that allows systems to **learn and improve from experience without being explicitly programmed**. Instead of fixed rules, ML models learn patterns from data.

Supervised	Labeled	Predict output	Spam detection
Unsupervised	Unlabeled	Find patterns	Customer clustering
Reinforcement	Environment feedback	Maximize reward	Game playing AI

Key ML concepts:

- **Training Data:** Dataset used to teach the model to recognise patterns.
- **Overfitting:** Model learns noise in training data and performs poorly on new data.
- **Underfitting:** Model is too simple to capture the underlying patterns.
- **Transfer Learning:** Reusing a pre-trained model for a new related task — reduces data requirements.

2.3 Deep Learning & Neural Networks

Deep Learning uses **multi-layered artificial neural networks** to model complex patterns in large datasets. Inspired by the human brain, each layer learns increasingly abstract features.

Neural Network	Nodes (neurons) arranged in input, hidden, and output layers — processes data through weighted connections.
CNN (Convolutional Neural Network)	Specialised for image/video recognition. Uses filters to detect spatial patterns.

RNN (Recurrent Neural Network)	Suited for sequential data (text, time-series). Connections form directed cycles.
GAN (Generative Adversarial Network)	Two networks — generator creates data, discriminator judges realism. Produces synthetic images, videos, etc.
LLM (Large Language Model)	Deep learning model trained on massive text. Understands and generates natural language. (e.g., GPT, Claude)

2.4 Key AI Fields

NLP (Natural Language Processing)	Enables computers to understand, interpret, and generate human language. Powers chatbots, translation, sentiment analysis.
Computer Vision	Trains computers to interpret visual data from images/videos. Used in medical imaging, self-driving cars, facial recognition.
Expert Systems	AI programs mimicking a human expert. Uses a knowledge base + inference engine. Early form of AI.
Robotics	AI-powered physical machines that sense, process, and act in the real world.
Generative AI	AI that creates new content (text, images, audio, code) based on training data. Examples: ChatGPT, DALL·E, Gemini.

2.5 AI Challenges & Ethics

- **Bias:** Systematic errors from biased training data or flawed model design — can cause unfair outcomes.
- **Explainability:** Many deep learning models are 'black boxes' — hard to understand why a decision was made.
- **Privacy:** AI systems often require vast amounts of personal data.
- **AI Alignment:** Ensuring AI systems behave in line with human values and intentions.
- **Job Displacement:** Automation threatens routine cognitive and manual jobs.

The Turing Test: Proposed by Alan Turing — if a machine's responses are indistinguishable from a human's in conversation, it can be considered intelligent.

SESSION 03 — Internet of Things (IoT) & Edge Computing

3.1 Defining IoT

The **Internet of Things (IoT)** refers to a network of interrelated devices embedded with sensors, software, and connectivity that enables them to collect and exchange data with other devices and cloud systems — without requiring human-to-human or human-to-computer interaction.

IoT (Wikipedia, 2008)	A wireless network between objects, self-configuring, like household appliances.
IoT (IoT 2008)	Technologies enabling the Internet to reach out into the real world of physical objects.
IoT in 2020	Things with identities and virtual personalities in smart spaces, using intelligent interfaces to connect in social, environmental, and user contexts.

3.2 The 'Any-X' Framework

IoT connects people and things ideally through **six dimensions**:

Anytime	Anyplace	Anything
Anyone	Any Path	Any Service

3.3 Characteristics of IoT

Intelligence	Knowledge extraction from generated data using analytics and AI.
Architecture	Hybrid architecture supporting many communication technologies.
Scalability	Must handle billions of simultaneous events and device connections.
Everything-as-a-Service	IoT resources consumed as on-demand services.
Complex System	Diverse, dynamically changing objects with time/space/location considerations.

3.4 IoT Layered Architecture

Perception Layer	Senses the physical environment	Sensors, RFID tags, cameras
Network Layer	Transmits collected data	Wi-Fi, 5G, Bluetooth, LoRaWAN
Middleware Layer	Data processing, storage, analysis	Cloud platforms, edge servers
Application Layer	Delivers service to end users	Smart home apps, dashboards

3.5 Key Technologies in IoT

RFID (Radio Frequency Identification)

- Integrates radios into almost all objects — one of the smallest but most critical IoT components.
- Composed of **one or more readers** and **tags**.
- **Tag structure:** a tiny microchip attached to an antenna — passive (no battery) or active (battery-powered).
- Used for tracking inventory, access control, and supply chain management.

Wireless Communication Technologies

Bluetooth	~10 m	Personal devices, wearables
Wi-Fi	~100 m	High-speed local networking
ZigBee / Z-Wave	10–100 m	Home automation, low power
LoRaWAN	2–15 km	Long-range, low-power IoT
5G	City-wide	Massive IoT, autonomous vehicles
RFID	cm–few m	Object identification, tracking

3.6 Sensor Networks (SNs)

Sensor Networks consist of a **large number of sensing nodes communicating wirelessly in a multi-hop fashion**. They are the backbone of IoT:

- **"SNs can exist without IoT, but IoT cannot exist without SNs."**
- Used for environmental monitoring, agriculture, medical care, and event detection.
- Middleware challenges: abstraction, data fusion, resource constraints, scalability, and QoS support.

3.7 Middleware & SOA

Middleware is the software layer between the Operating System and Application. It solves issues of **heterogeneity, interoperability, security, and dependability** in IoT systems.

Service Oriented Architecture (SOA): A middleware approach that allows software/hardware reuse without imposing specific technology — uses a layered model addressing Abstraction, Common Services, and Composition.

3.8 Edge Computing

Edge Computing processes data **at or near the source** (sensors, gateways) rather than sending it to a distant cloud data centre. It is essential for latency-sensitive IoT applications.

Reduced Latency	Real-time response for autonomous vehicles, industrial control
Bandwidth Saving	Only relevant data sent to cloud, reducing costs
Improved Privacy	Sensitive data processed locally, not exposed to cloud
Reliability	Continues operating even if cloud connection is lost

3.9 Open Challenges in IoT

Scalability	Billions of devices need unique addresses and data management strategies.
Standardization	No universal protocol — devices from different vendors often cannot communicate.
Big Data	IoT generates enormous data streams that are hard to store, process, and understand.
Security & Privacy	Larger attack surface, hard-to-update low-power sensors, surveillance risks.
Addressing	IPv6 needed to accommodate billions of uniquely addressed IoT devices.
QoS	Smart object traffic patterns are different — networks need Quality of Service guarantees.

3.10 IoT Applications

Smart Cities	Traffic management, smart lighting, waste monitoring, parking sensors
Agriculture	Soil moisture monitoring, automated irrigation, crop health sensors
Healthcare	Remote patient monitoring, wearables, smart hospitals, asset tracking
Industry (IIoT)	Predictive maintenance, automated assembly lines, quality control
Smart Home	Smart speakers, thermostats, security cameras, connected appliances
Logistics	Real-time fleet tracking, cold-chain monitoring, inventory management